Discrimination and Calibration of Machine-Learning Models for Auto-Insurance Claim Prediction: Does SMOTE Help?

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My (Chien-Kai) contribution: I led our team's investigation comparing supervised models for auto-insurance claim prediction. I was responsible for the logistic regression and XGBoost models, including the XGBoost hyperparameter search under five-fold cross-validation, and I contributed to the SMOTE implementation and one-hot encoding. I investigated why SMOTE distorts predicted probabilities (see Section 5.2) and directed the evaluation's use of the Brier score, calibration analysis, and paired bootstrap confidence intervals and significance tests reported throughout the paper, which my teammates helped implement in code. Our instructor taught us machine learning theory and models and gave us advice only occasionally.

Abstract

Insurers need claim probabilities, not only claim flags. Claims are rare events (6.4% of policies in the data used here), and oversampling methods such as SMOTE are widely applied to cope with this imbalance. We compare logistic regression, a decision tree, a random forest, and XGBoost on a public Kaggle auto-insurance dataset (58,592 policies, 42 predictors), with and without SMOTE, under a protocol designed to avoid test-set leakage. SMOTE is applied inside cross-validation training folds only, classification thresholds are chosen on out-of-fold predictions, and a stratified 20% test set is scored once. The models are compared on discrimination (ROC-AUC, PR-AUC) and probability quality (Brier score), with 95% bootstrap confidence intervals that are paired for differences between models and between scenarios.

XGBoost without SMOTE had the best discrimination (ROC-AUC 0.666, 95% CI 0.646–0.686; PR-AUC 0.114, 95% CI 0.101–0.128, against a base rate of 0.064), but discrimination was modest for every model. The effect of SMOTE depended on the model. It left the ranking ability of the decision tree statistically unchanged and that of logistic regression essentially unchanged (changes of about -0.002 in ROC-AUC and PR-AUC), reduced it for XGBoost, and improved it for the random forest, whose no-SMOTE baseline was the weakest. At the same time, SMOTE inflated the Brier score of logistic regression (0.059 to 0.143) and XGBoost (0.059 to 0.071). Where predicted probabilities matter, as in insurance pricing, models trained with SMOTE should not be used without recalibration, which was not evaluated here.

Keywords: claim prediction; class imbalance; SMOTE; probability calibration; Brier score; XGBoost; bootstrap.

1 Introduction

Insurers use claim-prediction models in two different ways. Ranking uses, such as triaging policies for underwriting review, need only a model that orders policies by risk. Pricing and reserving uses need predicted probabilities that match the true claim frequency. Both uses are made harder by the fact that claims are rare, since only about 6.4% of the policies in the data analyzed here have a claim.

Researchers commonly handle class imbalance with resampling, and SMOTE [CBHK02] is among the most widely used methods. Applied studies often judge such methods by recall or F1 at a fixed or tuned threshold. Three problems can arise in this practice. First, resampling changes the class proportions seen during training, so predicted probabilities no longer reflect real-world claim rates [vdGvSTVC22], and many classifiers are poorly calibrated even without resampling [NMC05]. Second, if thresholds are tuned on the test set, reported performance is optimistic. Third, because recall and F1 depend on a single threshold, they hide how well a model ranks policies, and for rare outcomes the precision–recall curve is more informative than the ROC curve alone [SR15].

To address these problems, we compare the models under a leakage-conscious protocol and with explicit uncertainty estimates. We address two research questions.

- **RQ1.** How do four common classifiers compare on discrimination and on probability quality for rare-event claim prediction?
- **RQ2.** Does SMOTE improve discrimination, and what does it do to probability quality?

This paper makes three contributions.

- an evaluation protocol that avoids test-set leakage and reports both discrimination and probability quality, with paired-bootstrap confidence intervals for differences between models and between SMOTE and no-SMOTE;
- evidence that the effect of SMOTE on discrimination depends on the model;
- evidence that SMOTE degrades probability quality for logistic regression and XGBoost.

2 Data

2.1 Source and target

The data are the public Kaggle "Car Insurance Claim Prediction" dataset (file `train.csv`). The data are anonymized: the insurer is not identified, and there is no calendar-time variable, explicit exposure measure, or claim amount. The target is `is_claim`, a binary indicator of whether the policyholder filed a claim in the next 6 months. The policy identifier was removed before modeling.

2.2 Predictors

Of the 42 predictors (Appendix A), 27 were stored as text in the source file and were one-hot encoded: 17 binary vehicle-equipment indicators and 10 other categorical vehicle and area attributes. The other 15 were stored as numbers and used as numeric variables. Nine of these are measurements such as policy tenure, policyholder age and vehicle age, and six are integer-coded vehicle attributes (`make`, `airbags`, `ncap_rating`, `cylinder`, `gear_box` and `displacement`).

2.3 Splitting and preprocessing

- A stratified 80/20 train–test split on `is_claim` (seed 123) was made before any preprocessing.
- All preprocessing rules were learned from the training set only: predictors with a single value were dropped, categorical variables were one-hot encoded (full rank), all-zero and exactly

collinear columns were removed, and every column was min–max scaled to $[0, 1]$ because SMOTE uses Euclidean distance.

- There were no missing values.

Table 1: Dataset summary.

Item	Value
Total policies	58,592
Training policies	46,875
Test policies	11,717
Model input columns after encoding	35
Claim rate, overall	6.40%
Claim rate, training	6.40%
Claim rate, test (no-skill PR-AUC baseline)	6.39%

3 Methods

3.1 Evaluation protocol

The same protocol was applied to every model, once without SMOTE and once with SMOTE (Table 2). The key design choices are that SMOTE never touches a validation fold, that the threshold is chosen without looking at the test set, and that the test set is scored exactly once.

Table 2: Evaluation protocol.

Step	What was done	Data used
1	Stratified 80/20 split on `is_claim` (seed 123). The test set is not touched again until Step 5.	Full data
2	Preprocessing rules learned on the training set only (constant columns dropped, one-hot encoding, removal of zero-variance and collinear columns, min–max scaling).	Training
3	Hyperparameter tuning by 5-fold cross-validation with identical folds for every model and both scenarios. Selection by out-of-fold PR-AUC (decision tree: 1-SE rule). In the SMOTE scenario, SMOTE is applied inside each CV training fold only; validation folds are never oversampled.	Training
4	Classification threshold chosen on the pooled out-of-fold predictions by maximizing the F_2 score over thresholds 0.01–0.90 (step 0.005).	Training (out-of-fold)
5	Final model refitted on the full training set (SMOTE-augmented in the SMOTE scenario). The test set is scored once; all reported metrics and bootstrap intervals come from these predictions.	Test

3.2 Models and tuning

Four models were fitted (Table 3): logistic regression, a CART decision tree, a random forest [Bre01] and gradient-boosted trees with XGBoost [CG16]. No model used class weights. SMOTE used the `smotefamily` package with $K = 5$ neighbors and seven synthetic rows per minority row, which raises the minority share of the training data from about 6% to roughly one third.

Table 3: Models and tuning.

Model	Implementation	Tuning and selection
Logistic regression	`glm`, logit link, all predictors, no penalty	None
Decision tree (CART)	`rpart` via `caret`; `minsplit`= 2	Complexity parameter $cp \in \{10^{-4}, 2\times10^{-4}, 5\times10^{-4}, 10^{-3}, 2\times 10^{-3}, 5\times10^{-3}, 10^{-2}\}$; 1-SE rule on CV PR-AUC
Random forest	`randomForest` via `caret`; 100 trees	`mtry`$\in \{0.5\sqrt{p}, \sqrt{p}, 2\sqrt{p}\}$ (rounded; p = number of columns); CV PR-AUC
XGBoost	`xgboost`, `binary:logistic`; $\eta = 0.1$; no early stopping	`nrounds`$\in \{100, 200, 300\}$ × `max_depth`$\in \{3, 4, 5\}$ × `min_child_weight`$\in \{1, 5, 10\}$ (27 combinations); $\gamma = 1$, `colsample_bytree`= 0.6, `subsample`= 0.8; CV PR-AUC

3.3 Metrics

- **Discrimination.** ROC-AUC, and PR-AUC computed as the area under the precision–recall curve (`PRROC` package). The no-skill PR-AUC equals the base rate, 0.064.
- **Probability quality.** The Brier score [Bri50], $\text{BS} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2$, the mean squared difference between predicted probability $\hat{p}_i$ and the 0/1 outcome y_i; lower is better. Always predicting the base rate gives a no-skill value of $0.0639 \times 0.9361 \approx 0.060$.
- **Operating point.** Precision, recall and F1 at the threshold chosen in Step 4, which maximizes $F_\beta = (1+\beta^2)\,PR/(\beta^2 P + R)$ with $\beta = 2$ (P = precision, R = recall), so recall is weighted more heavily than precision.

Accuracy is not reported: a classifier that never predicts a claim is 93.6% accurate.
After one-hot encoding, the data contained 100 columns from 42 predictors; `findLinearCombos` removed 65 exactly collinear columns, most of them vehicle specification and equipment variables (engine type, torque, power, dimensions, standard safety features) that are fully determined by the vehicle model, leaving 35 columns for modeling. This indicates the effective dimensionality of the vehicle-attribute variables is much lower than the raw variable count suggests.

3.4 Uncertainty quantification

Uncertainty was quantified with the nonparametric bootstrap [ET93]. The 11,717 test policies were resampled with replacement 1,000 times, using the same resamples for every model. Each metric was recomputed on each resample, and 95% percentile intervals were taken. Differences between models, and between SMOTE and no-SMOTE for the same model, were computed within each resample, so the intervals for differences are paired. These intervals reflect sampling variability of the test set only; they do not capture variation from other train–test splits, CV folds or random seeds.

3.5 Software

All analysis was done in R 4.5.1 using `caret`, `xgboost`, `randomForest`, `rpart`, `smotefamily`, `PRROC`, `MLmetrics`, `ggplot2`, `dplyr` and `tidyr`.

4 Results

4.1 Overall performance

Table 4 and Figure 1 show test-set performance for all eight model–scenario combinations. XGBoost without SMOTE ranked best on both discrimination metrics: ROC-AUC 0.666 and PR-AUC 0.114, about 1.8 times the base rate of 0.064. Every PR-AUC interval lies above the base rate and every ROC-AUC interval lies above 0.5, so all models carry some signal, but discrimination is modest throughout (Figure 2).

Without SMOTE, logistic regression and XGBoost have Brier scores (0.059) close to the no-skill value of 0.060: their probabilities are roughly calibrated but add little resolution beyond the base rate. For XGBoost the Brier skill score relative to the no-skill predictor is about 2% (a point estimate that was not tested).

Table 4: Test-set performance with 95% bootstrap confidence intervals in brackets.

Scenario	Model	ROC-AUC	PR-AUC	Brier score
No SMOTE	Logistic Regression	0.614 [0.594, 0.635]	0.094 [0.084, 0.106]	0.059 [0.055, 0.063]
No SMOTE	XGBoost	**0.666 [0.646, 0.686]**	**0.114 [0.101, 0.128]**	0.059 [0.055, 0.063]
No SMOTE	Decision Tree	0.596 [0.575, 0.619]	0.087 [0.079, 0.097]	0.073 [0.069, 0.078]
No SMOTE	Random Forest	0.569 [0.549, 0.590]	0.078 [0.071, 0.086]	0.062 [0.058, 0.066]
With SMOTE	Logistic Regression	0.612 [0.593, 0.633]	0.092 [0.083, 0.104]	0.143 [0.141, 0.144]
With SMOTE	XGBoost	0.647 [0.628, 0.666]	0.101 [0.091, 0.113]	0.071 [0.068, 0.074]
With SMOTE	Decision Tree	0.607 [0.586, 0.627]	0.090 [0.080, 0.101]	0.067 [0.064, 0.071]
With SMOTE	Random Forest	0.625 [0.605, 0.645]	0.093 [0.084, 0.104]	0.066 [0.062, 0.070]

No-skill references: ROC-AUC 0.5; PR-AUC 0.064 (test-set claim rate); Brier 0.060. Bold: best discrimination.

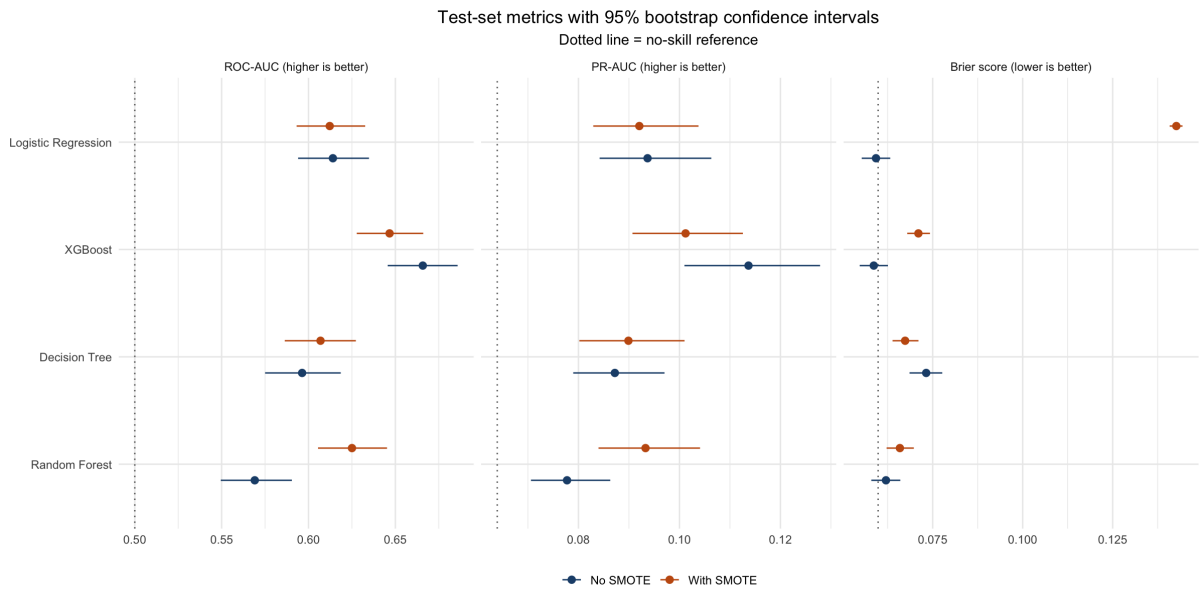


Figure 1: Test-set ROC-AUC, PR-AUC and Brier score with 95% bootstrap confidence intervals. Dotted lines are no-skill references.

4.2 Comparison of models

Without SMOTE, XGBoost outperformed each of the other three models in ROC-AUC and PR-AUC, with paired-bootstrap intervals that exclude zero (Table 5, Figure 3). The ROC-AUC

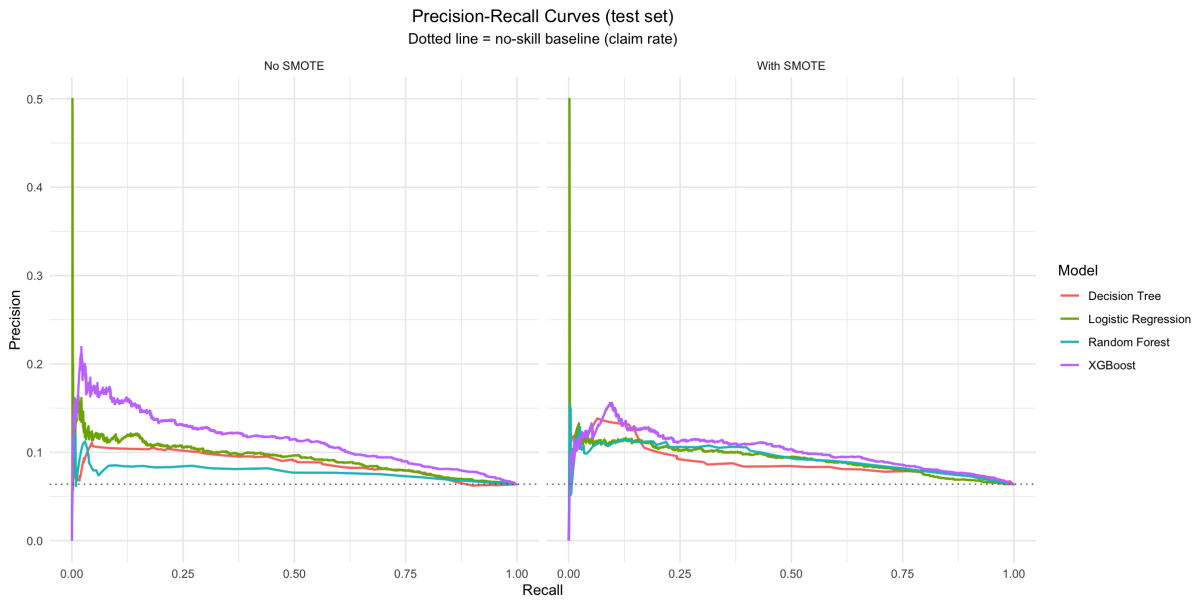


Figure 2: Precision–recall curves on the test set for each model, without SMOTE (left) and with SMOTE (right). The dotted line is the no-skill baseline (claim rate 0.064).

advantage was 0.052 over logistic regression, 0.070 over the decision tree and 0.097 over the random forest. XGBoost also had a lower Brier score than every other model, but the difference from logistic regression (0.0006) is negligible in practice even though its interval excludes zero.

Two caveats apply. The random forest was tuned over a much smaller search space than XGBoost (only `mtry`, with 100 trees), so part of its gap may reflect tuning effort rather than the method. And all comparisons rest on a single train–test split.

Table 5: XGBoost minus each other model, without SMOTE. Paired-bootstrap 95% CI in brackets; all intervals exclude 0.

Comparison	Δ ROC-AUC	Δ PR-AUC	Δ Brier score
XGBoost − Logistic Regression	+0.0518 [0.0353, 0.0675]	+0.0200 [0.0082, 0.0320]	−0.0006 [−0.0008, −0.0004]
XGBoost − Decision Tree	+0.0695 [0.0491, 0.0899]	+0.0265 [0.0158, 0.0382]	−0.0146 [−0.0168, −0.0123]
XGBoost − Random Forest	+0.0967 [0.0732, 0.1181]	+0.0359 [0.0247, 0.0485]	−0.0034 [−0.0039, −0.0028]

A positive difference in ROC-AUC or PR-AUC, or a negative difference in Brier score, favors XGBoost.

4.3 Effect of SMOTE

Table 6 and Figure 4 give the effect of SMOTE for each model as the paired difference (With SMOTE − No SMOTE). The effect was not uniform:

- **Logistic regression.** Ranking was essentially unchanged (ΔROC-AUC −0.002, interval includes 0; ΔPR-AUC −0.002, interval barely excludes 0), but the Brier score rose by 0.083, from 0.059 to 0.143.
- **XGBoost.** Both ranking and probability quality worsened: ΔROC-AUC −0.019, ΔPR-AUC −0.0125, ΔBrier +0.012.
- **Decision tree.** Ranking was unchanged and the Brier score was lower (−0.006). However, the no-SMOTE tree had a Brier score (0.073) worse than the no-skill value, which points to

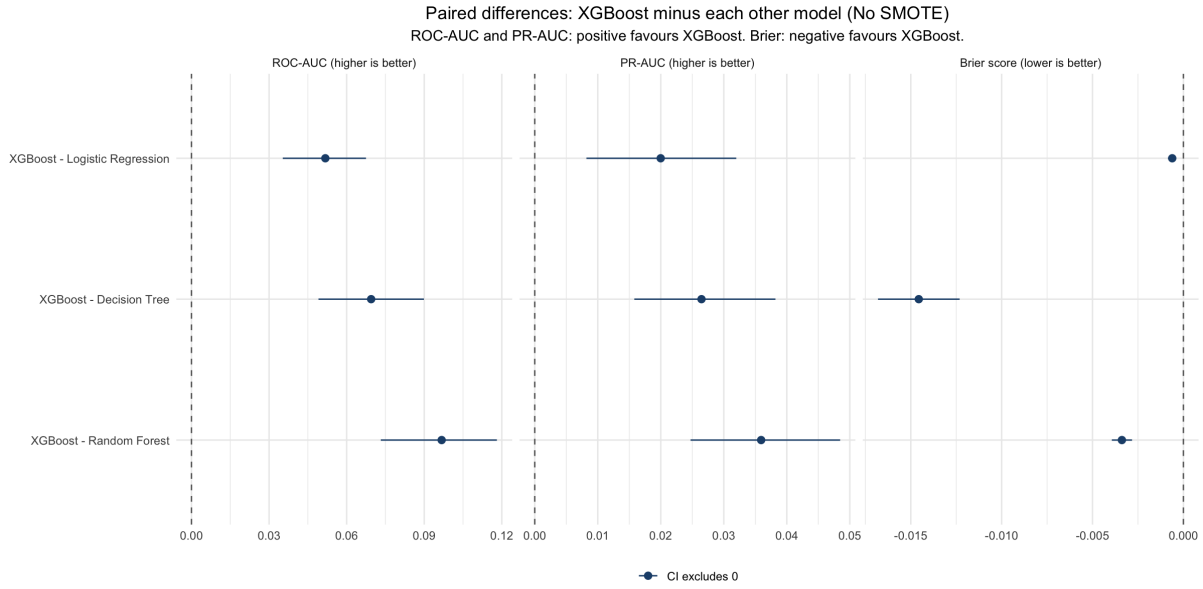


Figure 3: Paired differences between XGBoost and each other model without SMOTE, with 95% bootstrap intervals.

overfitted leaf probabilities, so this should not be read as SMOTE improving calibration as such.

- **Random forest.** Ranking improved (ΔROC-AUC +0.056, ΔPR-AUC +0.016) while the Brier score rose slightly (+0.004). The no-SMOTE random forest had the weakest discrimination of all models (ROC-AUC 0.569), so part of this gain may come from a poorly tuned baseline rather than from SMOTE itself.

Figure 5 shows the calibration pattern behind these Brier score changes.

Table 6: Effect of SMOTE (With SMOTE − No SMOTE). Paired-bootstrap 95% CI in brackets; bold: interval excludes 0.

Model	Δ ROC-AUC	Δ PR-AUC	Δ Brier score
Logistic Regression	−0.0018 [−0.0046, 0.0009]	**−0.0016 [−0.0037, −0.0001]**	**+0.0834 [0.0805, 0.0863]**
XGBoost	**−0.0191 [−0.0324, −0.0064]**	**−0.0125 [−0.0207, −0.0057]**	**+0.0124 [0.0111, 0.0136]**
Decision Tree	+0.0106 [−0.0158, 0.0348]	+0.0027 [−0.0069, 0.0113]	**−0.0059 [−0.0083, −0.0034]**
Random Forest	**+0.0560 [0.0346, 0.0778]**	**+0.0155 [0.0076, 0.0233]**	**+0.0039 [0.0027, 0.0050]**

For Δ ROC-AUC and Δ PR-AUC, positive means SMOTE is better; for Δ Brier score, positive means SMOTE is worse.

Tables 5 and 6 contain 21 paired comparisons in total, and no correction for multiple comparisons was applied. Because such a correction would widen every interval, results whose intervals lie close to zero should be read cautiously. These are the effect of SMOTE on PR-AUC for logistic regression (upper limit −0.0001), XGBoost (upper limit −0.006) and the random forest (lower limit +0.008), the effect of SMOTE on the ROC-AUC of XGBoost (upper limit −0.006), and the PR-AUC advantage of XGBoost over logistic regression (lower limit +0.008). The remaining differences are less sensitive to such a correction.

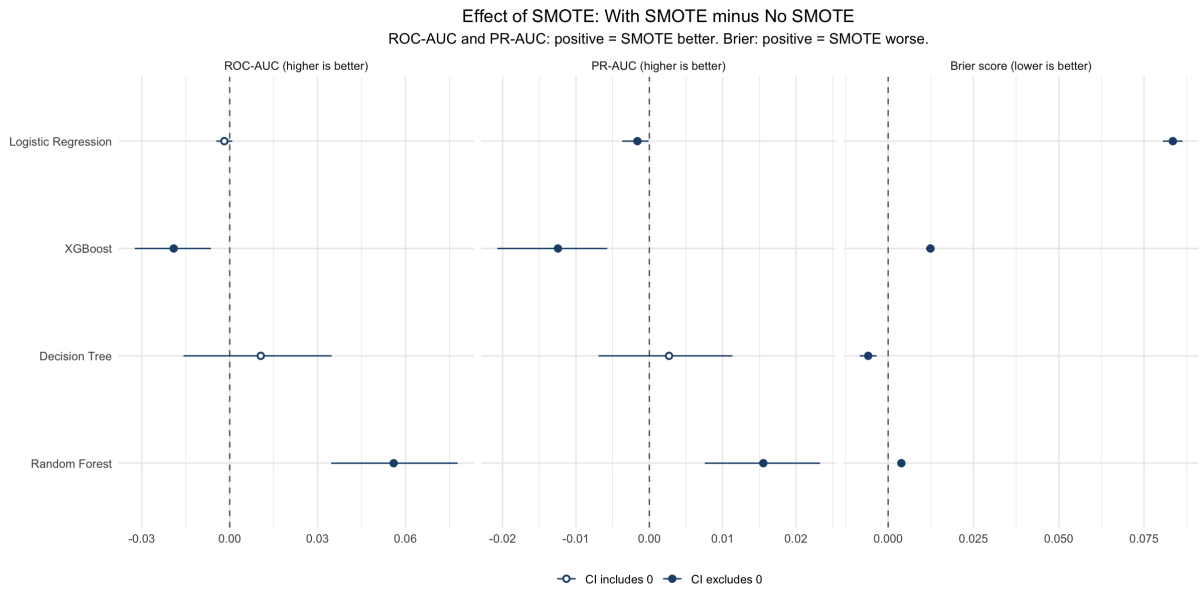


Figure 4: Effect of SMOTE (With SMOTE minus No SMOTE) on ROC-AUC, PR-AUC and Brier score, with 95% paired-bootstrap intervals.

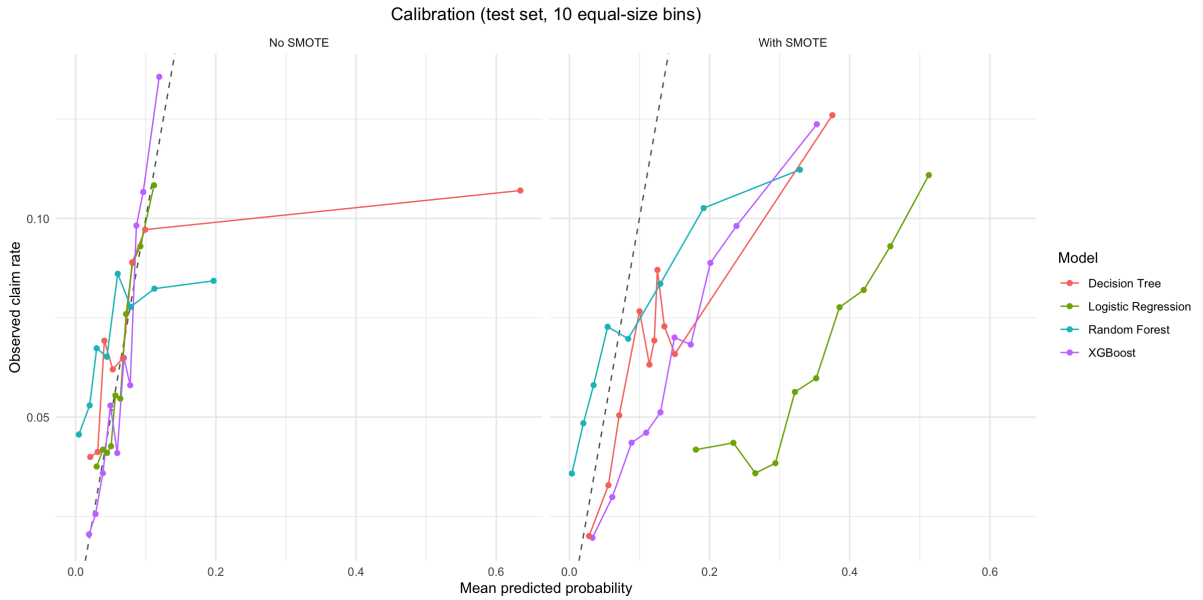


Figure 5: Calibration on the test set (ten equal-size bins of predicted probability). Points on the dashed diagonal indicate predicted probabilities that match observed claim rates. The SMOTE models generally lie below the diagonal, indicating systematic overprediction of claim probabilities, most strongly for logistic regression. Without SMOTE, logistic regression and XGBoost lie much closer to the diagonal, whereas the decision tree's top bin (mean predicted probability 0.63, observed claim rate about 0.11) and the flat random-forest curve show clear miscalibration.

4.4 Operating points

Thresholds were chosen on out-of-fold training predictions by maximizing F_2 (Figure 6). Table 7 reports the resulting operating points on the test set. Every model reaches a recall of 0.64–0.92 only by flagging a large share of policies as likely claims. For XGBoost without SMOTE, precision is 0.085 at a recall of 0.79, which corresponds to flagging roughly 59% of all policies. Across all configurations, precision is 0.063–0.088, about 1.0–1.4 times the base rate.

Without SMOTE, the models rarely output a probability above 0.5, so a fixed 0.5 threshold gives a recall of exactly 0.000 for three of the four models. This is why recall reported at an arbitrary or test-tuned threshold is a poor summary of model quality.

Table 7: Operating points on the test set at the threshold chosen on out-of-fold training predictions (F_2).

Scenario	Model	Threshold	Precision	Recall	F1	Flagged share	Recall at 0.5
No SMOTE	Logistic Regression	0.0500	0.0765	0.7864	0.1394	66%	0.0000
No SMOTE	XGBoost	0.0550	0.0847	0.7864	0.1529	59%	0.0000
No SMOTE	Decision Tree	0.0200	0.0629	0.9226	0.1177	94%	0.0174
No SMOTE	Random Forest	0.0150	0.0717	0.7877	0.1314	70%	0.0000
With SMOTE	Logistic Regression	0.2900	0.0767	0.7944	0.1399	66%	0.1055
With SMOTE	XGBoost	0.1150	0.0814	0.7957	0.1477	62%	0.0080
With SMOTE	Decision Tree	0.0950	0.0793	0.8037	0.1444	65%	0.0280
With SMOTE	Random Forest	0.0450	0.0883	0.6395	0.1551	46%	0.0040

Flagged share = recall × base rate ÷ precision (derived, approximate): the fraction of test policies predicted to have a claim.

4.5 Which variables carry the signal

To describe which predictors are associated with claims, a logistic regression was fitted to the original, non-oversampled training data. The p-values from SMOTE-augmented data are not valid, because synthetic rows are not independent observations and inflate the effective sample size.

Variables significant at the 5% level are listed in Table 8, and random-forest variable importance is shown in Figure 7. These are associations in observational data, not causal effects.

Policy tenure and vehicle age were consistently among the most important predictors. The random-forest rankings also highlighted policyholder age, particularly after SMOTE, while the logistic regression identified policy tenure and vehicle age as significant predictors and additionally found associations with two area-cluster categories. Thus, the models show broad agreement on policy and vehicle characteristics, although their detailed rankings differ.

No adjustment was made for multiple testing, so the two area-cluster results in particular should be read cautiously.

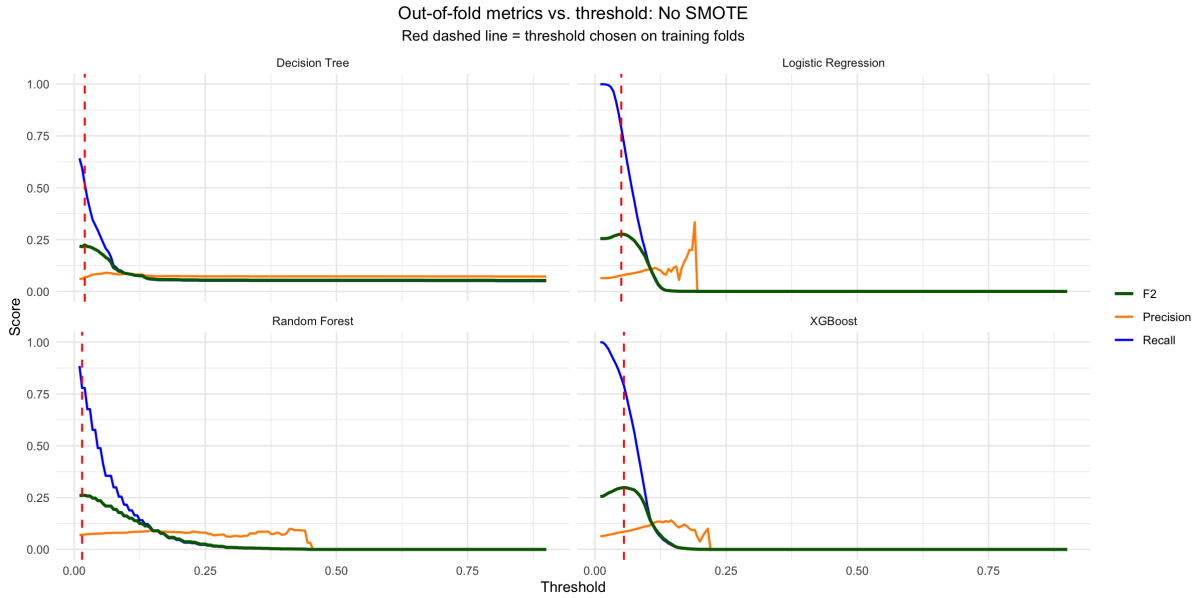
Table 8: Predictors significant at the 5% level in the logistic regression fitted to the original training data. All predictors were min–max scaled to $[0, 1]$, so each coefficient is the change in log-odds across the training-set range of that predictor. Area-cluster coefficients are relative to the omitted reference level.

Variable	Coefficient estimate	p-value
policy_tenure	1.1530090	7.135037e-60
age_of_car	−3.7318713	5.545833e-20
area_cluster.C18	0.7160344	6.847574e-03
area_cluster.C4	0.4153285	4.410461e-02

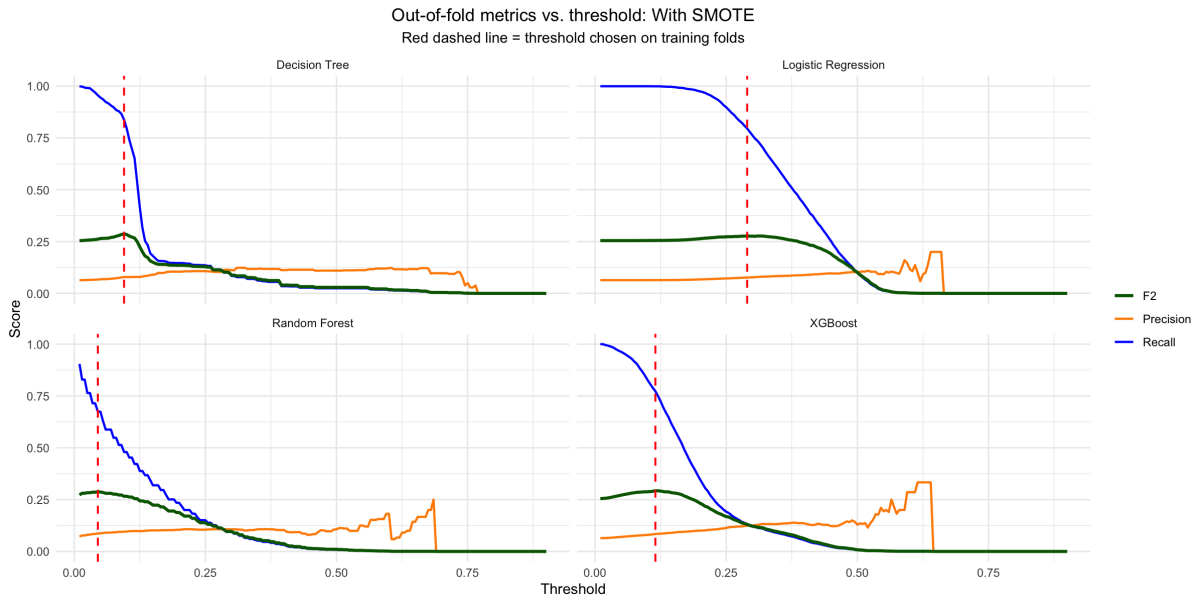
5 Discussion

5.1 Main findings

Three findings stand out. First, XGBoost without SMOTE had the best discrimination, but the absolute level was low (ROC-AUC 0.67). Second, SMOTE left discrimination statistically



(a) Without SMOTE



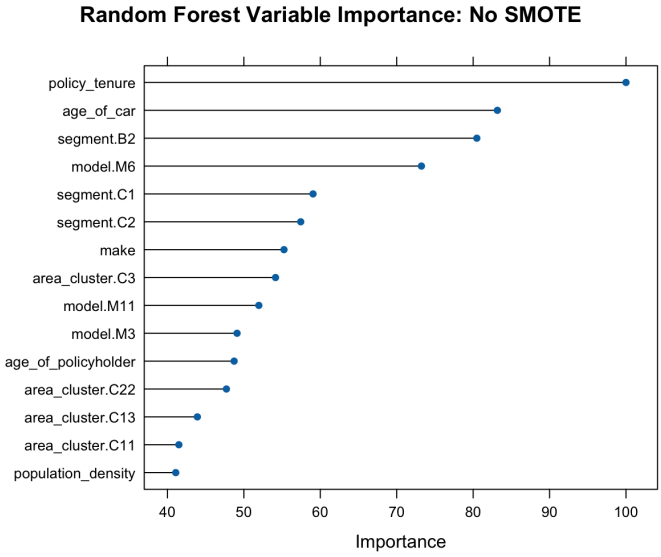
(b) With SMOTE

Figure 6: Out-of-fold precision, recall and F_2 versus classification threshold for each model. Red dashed lines mark the thresholds selected on the training folds.

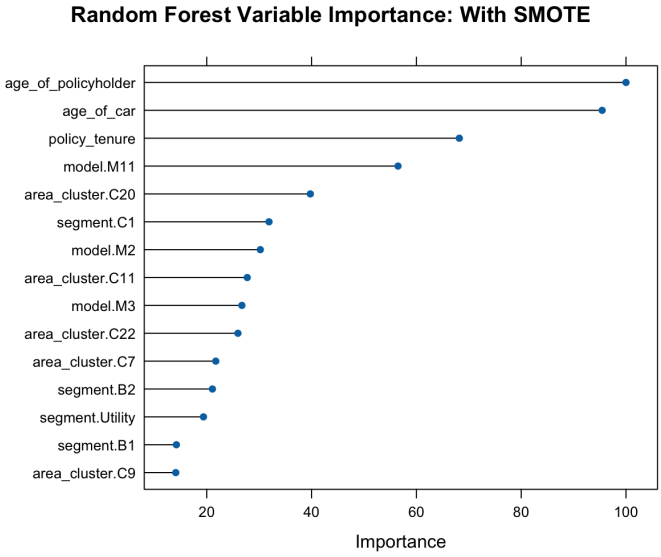
unchanged for the decision tree and essentially unchanged for logistic regression, reduced it for XGBoost, and improved it for the random forest, whose no-SMOTE baseline was weak. Third, SMOTE substantially worsened the Brier score of logistic regression and XGBoost.

5.2 Why SMOTE may distort predicted probabilities

With the settings used here, SMOTE raises the share of positive cases in the training data from about 6% to roughly one third. A model fitted to such data learns a claim prior several times too high, so its predicted probabilities are shifted upward relative to the real claim rate, and the Brier score penalizes this. The calibration plot (Figure 5) is consistent with this account, since the SMOTE models generally lie below the diagonal. The unchanged ROC-AUC of logistic



(a) Without SMOTE



(b) With SMOTE

Figure 7: Random-forest variable importance (top 15 variables).

regression fits the same explanation, because a shift in the prior mainly moves predictions up without reordering policies. The account does not apply equally to every model. The decision tree's Brier score fell after SMOTE, which Section 4.3 attributes to overfitted leaf probabilities in the no-SMOTE tree, and for tree-based models oversampling also changes the fitted structure, which can move discrimination in either direction, as seen here. A prior-shift correction was not tested, so this remains an interpretation rather than a demonstrated mechanism.

5.3 Implications for insurance use

For rank-based uses, a shifted probability scale may be acceptable, and the choice among resampling, class weights and no correction should be made on discrimination metrics together with their uncertainty. Class weights were not evaluated here. For pricing, reserving, or any use that needs the probability itself, an oversampled model should not be used without recalibration

on data with the true claim rate (for example Platt scaling, isotonic regression, or a prior-shift correction); recalibration was not evaluated here. Thresholds should reflect the costs of false positives and false negatives, estimated on validation data, rather than being fixed at 0.5 or chosen by recall alone. The thresholds used in this paper were chosen by maximizing F_2 and were not derived from assumed costs.

5.4 Modest signal

The signal in these data is weak. Even the best model's Brier score is only about 2% below the no-skill value, and its 95% interval (0.055 to 0.063) includes that value, although the difference itself was not tested. High recall is obtained only by flagging a large share of policies. This plausibly reflects the fact that the dataset is anonymized and contains no driver behavior, claims history or explicit exposure information. The results should not be read as evidence that any of these models is ready for underwriting use.

6 Limitations

- **Single split and seed.** The confidence intervals capture test-set sampling variability but not variation from other splits, CV folds or random seeds. Repeated cross-validation or several seeds would be a stronger design.
- **Multiple comparisons.** Twenty-one paired comparisons were made without correction.
- **Calibration assessed descriptively.** Probability quality was measured with the Brier score, which combines calibration and discrimination, and with a decile plot (Figure 5) that has no confidence bands. Each decile holds roughly 1,170 test policies and about 75 claims, so an observed rate has a standard error near 0.007, and small fluctuations in the plot should not be interpreted. No calibration slope or intercept was estimated.
- **Unequal tuning.** The random forest and decision tree were tuned over narrower spaces than XGBoost, and the XGBoost grid has only 27 combinations. The finding that SMOTE helps the random forest may reflect a weak baseline.
- **Reuse of out-of-fold predictions.** The same out-of-fold predictions were used for hyperparameter selection and for threshold selection, which may make training-side numbers slightly optimistic; the test set is unaffected.
- **Data.** The dataset is anonymized and public, with no time index, explicit exposure measure or claim amount. The target is a binary claim indicator over a fixed window, so the results say nothing about claim frequency per exposure or severity.
- **Selections at grid boundaries.** Several selected hyperparameters lie at the edge of their grid (Tables 3 and B1), for example `nrounds`= 100 and `max_depth`= 3 for XGBoost without SMOTE, `max_depth`= 5 and `min_child_weight`= 10 for XGBoost with SMOTE, and `mtry`= 12 and 3 for the random forest without and with SMOTE, so wider grids might change the results.
- **Integer-coded predictors.** Six integer-valued vehicle attributes (`make`, `airbags`, `ncap_rating`, `cylinder`, `gear_box` and `displacement`) were used as numeric variables rather than one-hot encoded. For counts, ratings and engine size this can be defensible, but it imposes a linear effect on the logit in the logistic regression. The case of `make` (vehicle manufacturer, five categories) is the most questionable because its codes are nominal. It was not significant in the logistic regression (Table 8) but ranked seventh in the random-forest importance without SMOTE (Figure 7(a)), so the effect of this choice on the reported results was not tested and may not be negligible.

7 Conclusion and future work

On a public auto-insurance dataset with a 6.4% claim rate, XGBoost without SMOTE gave the best discrimination among four common classifiers, although the signal was modest for every model. SMOTE had no consistent benefit for discrimination and inflated the Brier score of logistic regression and XGBoost. When predicted probabilities matter, as they do in insurance pricing, oversampling should be accompanied by recalibration, and models should be compared on discrimination and probability quality with uncertainty estimates rather than on recall at a tuned threshold.
Future work includes:

- evaluating recalibration (Platt scaling, isotonic regression) after SMOTE, with reliability diagrams and Brier-score decomposition;
- repeated cross-validation with several seeds, and a regularized GLM or GAM as a stronger interpretable baseline;
- cost-sensitive threshold selection under assumed loss ratios;
- validation on other insurance datasets, ideally with a time index for out-of-time testing.

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A Predictors

Table A1: The 42 predictors.

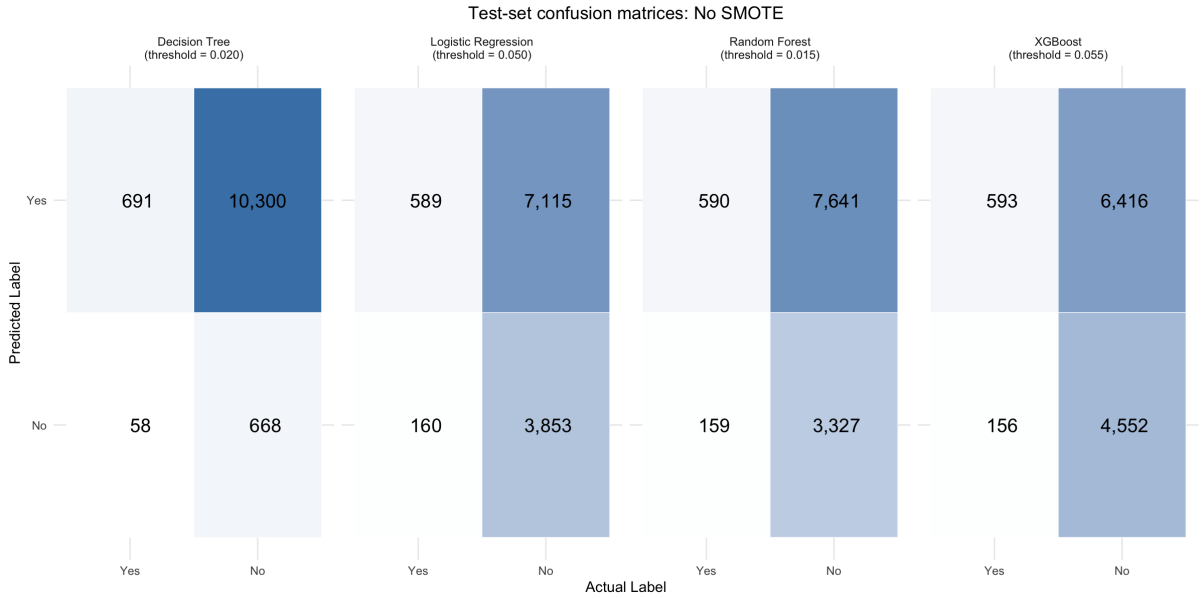
Group	Variables
Binary, safety assistance (8)	`is_esc`, `is_brake_assist`, `is_tpms`, `is_ecw`, `is_speed_alert`, `is_adjustable_steering`, `is_power_steering`, `is_driver_seat_height_adjustable`
Binary, parking and visibility (7)	`is_parking_sensors`, `is_parking_camera`, `is_front_fog_lights`, `is_rear_window_wiper`, `is_rear_window_washer`, `is_rear_window_defogger`, `is_day_night_rear_view_mirror`
Binary, locking (2)	`is_power_door_locks`, `is_central_locking`
Categorical, area and class (2)	`area_cluster`, `segment`
Categorical, make and model (2)	`make`†, `model`
Categorical, engine and specification (6)	`fuel_type`, `max_torque`, `max_power`, `engine_type`, `displacement`†, `cylinder`†
Categorical, transmission and control (3)	`transmission_type`, `gear_box`†, `steering_type`
Categorical, safety equipment and rating (3)	`airbags`†, `rear_brakes_type`, `ncap_rating`†
Numeric, policy (2)	`policy_tenure`, `age_of_policyholder`
Numeric, vehicle and area (7)	`age_of_car`, `population_density`, `turning_radius`, `length`, `width`, `height`, `gross_weight`

† Stored as integers in the source file and used as numeric variables, not one-hot encoded.

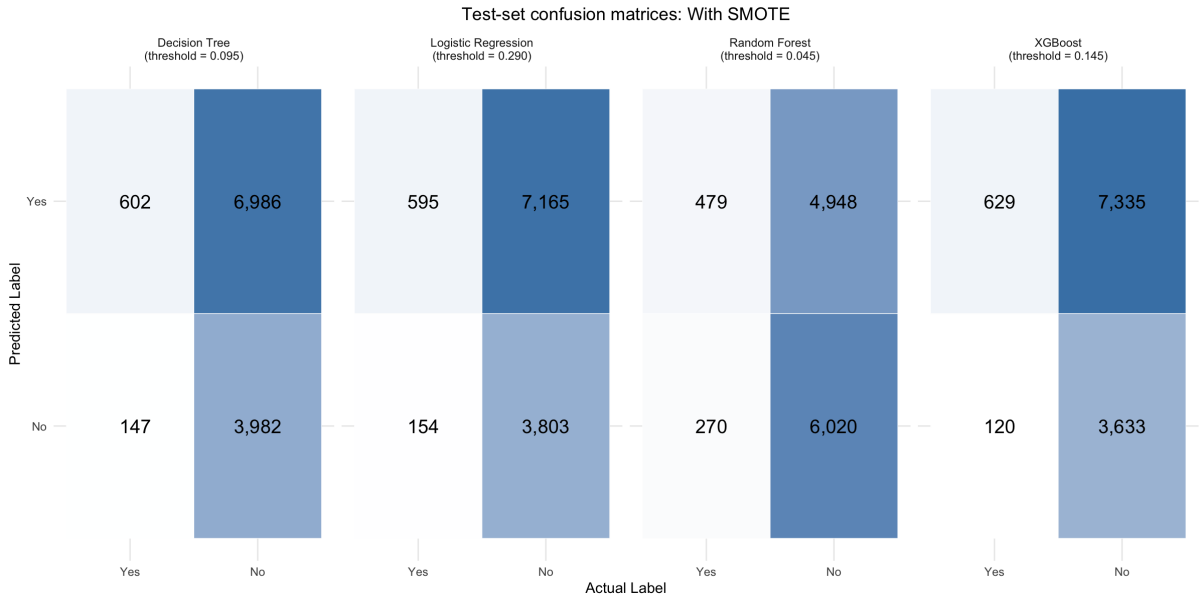
B Selected hyperparameters and confusion matrices

Table B1: Hyperparameters selected by cross-validation, by model and scenario.

Scenario	Model	Selected values
No SMOTE	Logistic Regression	(no tuning)
No SMOTE	XGBoost	nrounds = 100, max_depth = 3, min_child_weight = 5
No SMOTE	Decision Tree	cp = 0.0002
No SMOTE	Random Forest	mtry = 12
With SMOTE	Logistic Regression	(no tuning)
With SMOTE	XGBoost	nrounds = 200, max_depth = 5, min_child_weight = 10
With SMOTE	Decision Tree	cp = 0.0005
With SMOTE	Random Forest	mtry = 3



(a) Without SMOTE



(b) With SMOTE

Figure B1: Test-set confusion matrices at the tuned thresholds.